

KhatPaani – System Execution & Architecture Document

KhatPaani is a smart water allocation and reservoir monitoring system designed for agricultural villages. The system automates data collection, reservoir estimation, irrigation demand calculation, fairness-based allocation, and scarcity forecasting using satellite imagery, APIs, and optimization algorithms.

1. Execution Flow

- 1. Fetch weather and satellite data
- 2. Estimate reservoir water storage
- 3. Calculate crop-wise water demand
- 4. Optimize fair water allocation
- 5. Generate scarcity forecasts
- 6. Deliver dashboard and SMS outputs

2. System Architecture

The system follows a hybrid Node.js + Python architecture. Node.js handles API coordination, scheduling, database communication, and dashboard integration, while Python modules process satellite imagery, forecasting, and optimization tasks.

Component	Purpose
NASA POWER API	Weather data acquisition
Sentinel-1 & Sentinel-2	Reservoir and water body analysis
Node.js Scheduler	Daily automated execution
Python Processing Modules	Forecasting and optimization
MongoDB	Data storage and allocation logs
Dashboard + SMS	Farmer and committee communication

3. Technical Workflow

The proposed pipeline operates on a daily execution cycle. Initially, the system fetches weather data through the NASA POWER API and satellite imagery through the Google Earth Engine API. The reservoir surface area is estimated using NDWI-based image processing, and water volume is calculated using calibrated depth-area relationships. Crop water demand is then estimated using the FAO-56 crop coefficient methodology, considering crop type, growth stage, soil characteristics, and irrigation efficiency. After estimating total demand, the allocation

module distributes water fairly among farms while ensuring conservation during scarcity conditions. Finally, forecasting modules estimate reservoir stress for future periods using weather predictions and historical variability patterns. Results are delivered through dashboards and SMS notifications.

4. Project Execution Steps

- Step 1:** Initialize farm and reservoir data
- Step 2:** Schedule automated daily execution using node-cron
- Step 3:** Fetch weather data from NASA POWER API
- Step 4:** Process Sentinel imagery using NDWI
- Step 5:** Estimate reservoir volume
- Step 6:** Compute crop water demand using FAO-56
- Step 7:** Run allocation optimization algorithm
- Step 8:** Generate scarcity forecasts
- Step 9:** Store results in MongoDB
- Step 10:** Deliver outputs through dashboard and SMS

5. Technology Stack

Layer	Technology
Frontend	React / Streamlit
Backend	Node.js, Express
Database	MongoDB
Processing	Python
Scheduling	node-cron
APIs	NASA POWER API, Google Earth Engine
Satellite Data	Sentinel-1, Sentinel-2

6. Conclusion

KhatPaani provides an automated and scalable approach for intelligent agricultural water management. By combining satellite imagery, weather APIs, predictive forecasting, and fairness-aware allocation, the system aims to improve irrigation efficiency, reduce water stress, and support sustainable governance in rural agricultural communities.